

## Assignment-1

### 1 What is Artificial Intelligence?

Artificial Intelligence (AI) is a technology that enables machines to think, learn, and make decisions like humans.

Some examples of artificial intelligence are : chatgpt , gemini , siri , claude , etc

### 2 Difference between AI, Machine Learning, Generative AI, and Agentic AI.

**Artificial Intelligence (AI):** AI is the ability of machines to perform tasks that normally require human intelligence.

**Example:** Siri.

**Machine Learning (ML):** Machine Learning is a branch of AI that enables computers to learn from data and improve automatically.

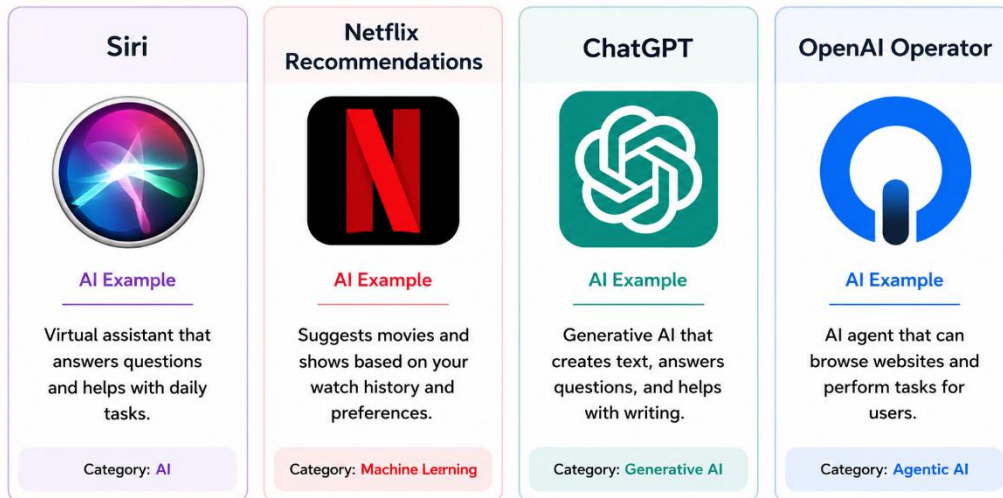
**Example:** Netflix recommendations.

**Generative AI:** Generative AI creates new content such as text, images, audio, or videos based on user prompts.

**Example:** Chatgpt .

**Agentic AI:** Agentic AI can make decisions, plan actions, and complete tasks with minimal human intervention.

**Example:** OpenAi operator.



### 3 Explain any three open-source AI models available on Hugging Face.

Any three open-source AI models available on hugging face are :

**Llama** – A powerful language model for text generation.

**Mistral** – An efficient model for chat and reasoning tasks.

**Stable Diffusion** – A model that generates images from text prompts.

### 3 Open-Source AI Models on Hugging Face



**1 Llama 3**  
(by Meta)

A powerful open-source language model for text generation, chat, and reasoning tasks.

**Example Use:**  
Chatbots, content generation, question answering.

Llama 3

What is the capital of France?

The capital of France is Paris.

**2 Mistral 7B**  
(by Mistral AI)

An efficient and high-performing language model for chat, code generation, and reasoning.

**Example Use:**  
Chat assistants, coding help, document summarization.

Mistral 7B

Write a Python function to add two numbers.

```
def add(a, b):
    return a + b
```

**3 Stable Diffusion XL**  
(by Stability AI)

An open-source text-to-image generation model that creates high-quality images from prompts.

**Example Use:**  
Image generation, concept art, design, marketing visuals.

Prompt:

A futuristic city at sunset with flying cars

💡 All these models are open-source and available on [Hugging Face Hub](#).

## 4 What is the difference between traditional programming and AI-assisted development?

### Traditional Programming:

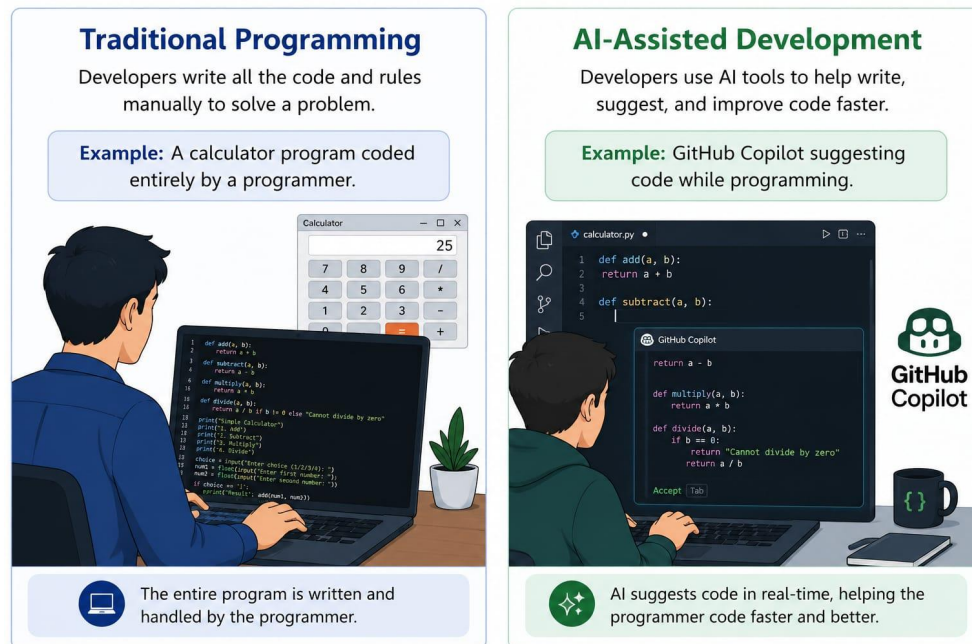
Developers write all the code and rules manually to solve a problem.

**Example:** A calculator program coded entirely by a programmer.

### AI-Assisted Development:

Developers use AI tools to help write, suggest, and improve code faster.

**Example:** GitHub Copilot suggesting code while programming.



## 5 Explain AI hallucinations with one example.

AI hallucination occurs when an AI generates false or made-up information and presents it as true.

### Examples

**Fake Book** AI says a book exists **and** gives an author name, but the book was never published.

**Wrong Historical Fact** :AI says the first moon landing happened in 1972 instead of 1969

**Fake Reference** : AI generates research paper citations that do not actually exist.

**Incorrect Biography** :AI claims a famous person won an award they never received.

**Made-up Website** : AI provides a website link that does not exist.

**Wrong Programming Answer** : AI suggests a function or library that is not available in the        programming language.

**Fake News Event** AI describes an event as real even though it never happened.

**Wrong Movie Information** : AI says a movie was directed by the wrong director.